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Classification of dividing and non-dividing cells in *Allium cepa* assay using YOLOv8

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Abstract

The *Allium cepa* assay is a valuable tool for evaluating natural products as potential anticancer agents by monitoring mitosis progression in onion root tips. However, the traditional method of identifying and classifying cells into various mitotic stages, and categorizing them as either dividing or non-dividing cells to calculate the mitotic index, can be laborious and prone to mistakes. This study aims to streamline the process of computing for the mitotic index using a convolutional neural network to identify and classify cells into different mitotic stages, and categorize them as dividing and non-dividing cells. A dataset of 1548 high-quality images of onion root tip cells displaying different stages of mitosis was used. A pretrained YOLOv8 model achieved an 88.33% mean average precision and 83% recall. Experiment results show that this approach can reduce the time-consuming aspects of the assay and potentially accelerate laboratory experiments performed in the search for novel anticancer drugs.

Keywords: Convolutional Neural Network; YOLO; Mitotic index; Mitosis; Allium cepa

1. Introduction

As cancer cells continue to evolve and develop multidrug resistance towards conventional anticancer treatments, recurrent tumor occurences becomes more prevalent [16]. Therefore, it is critical to discover new and potent anticancer medications [13]. Cancer is a genetic disease that stems from the changes in genes that regulate the function of our cells especially their growth and division. It manifests as an uncontrolled development of cells that can affect various parts of the body [7].

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In the search for anticancer drugs, thousands of potent natural products undergo phytochemical screenings, and antimitotic tests are performed to evaluate their potential as anticancer agents ([8]; [2]; [11]; [10]). A substance showing an antimitotic activity will be subjected to further study as a potential anticancer agent [4]. The *Allium cepa* test is an antimitotic test that involves exposing onion root tips to the candidate substance and observing its effects on the progression of mitosis.

One of the significant but tedious phases of the *Allium cepa* assay is the counting of the dividing and non-dividing cells to determine the mitotic index. Non-dividing cells refer to cells in *interphase* while dividing cells consist of cells in *prophase*, *metaphase*, *anaphase*, and *telophase*. The mitotic index is the ratio of the number of dividing cells and total number of cells [3]. It is a measure of similarity between a substance under investigation and a verified anticancer medication. The traditional method of manually calculating the mitotic index is laborious and prone to mistakes [5].

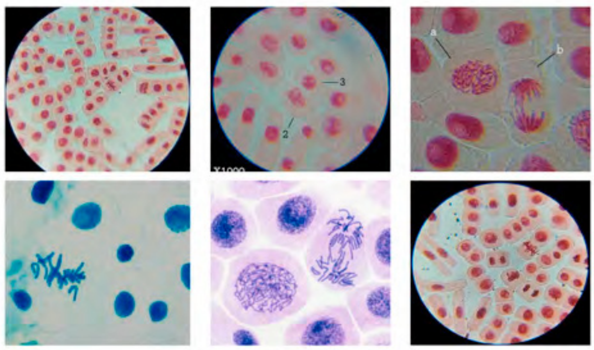
Convolutional Neural Networks (CNN) have been used to classify dividing cells and non-dividing cells for breast cancer ([1]; [9]; [6]; [12]; [15]), and drosophila pupal wing epithelium [14]. This is done due to the fact that it is effective at detecting specific features in a complex and dense field, such as images of cells. CNNs have also enabled the exploration of cell division within a group of cells, and in studying how one cell dividing might affect the behavior of nearby cells.

This study presents a Convolutional Neural Network based model for the real-time detection and classification of the different stages of mitosis of *Allium cepa* (onion) cells into dividing and non-dividing cells. The findings of this study can streamline the process of calculating the mitotic index in laboratory experiments that employ the *Allium cepa* mitotic test.

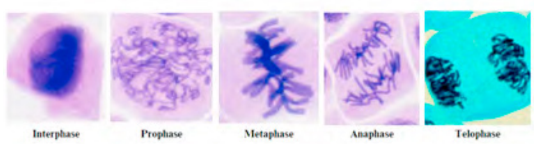
2. Materials and methods

2.1. Data Collection

The study utilizes high-quality images of squashed and stained onion root tip cells undergoing mitosis. The primary dataset comprises 729 high-quality images, including frames extracted from high-resolution microscope videos capturing onion root tip cells. These images originated from experiments conducted by Professor Durbal Goswami, M.Sc. (Zoology) at Tripura University, India. Supplementary to this primary collection, 16 additional images were sourced from various internet platforms to complement the dataset. Sample images are presented in Figure 1a and stages of mitosis are illustrated in 1b.



(a) Sample onion root tip cells undergoing mitosis

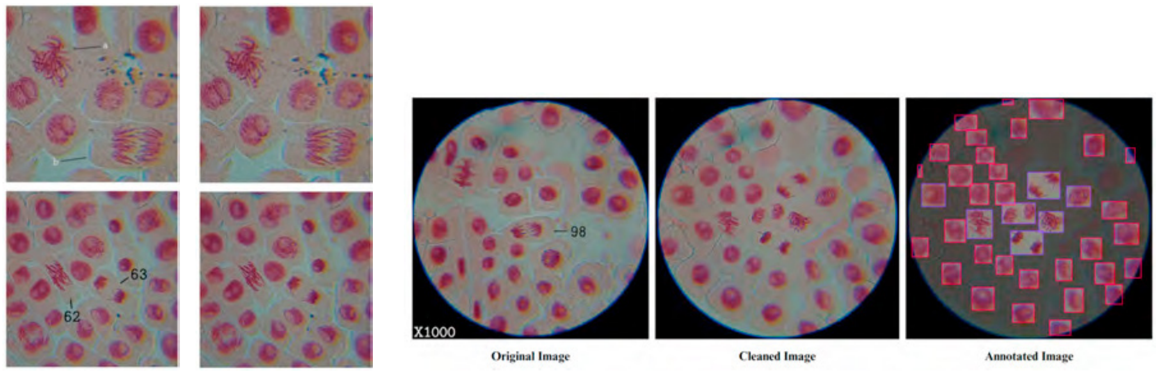


(b) Stages of mitosis

Fig. 1: Onion root tip cells

2.2. Image Preprocessing

Some images in the dataset contained artificial marks that could potentially influence the learning process of the model. To maintain focus on cell analysis, these artificial marks were manually removed from the images. This refinement process reduced the dataset to 644 images. Sample images before and after the artificial marks were removed are shown in Figures 2a and 2b .

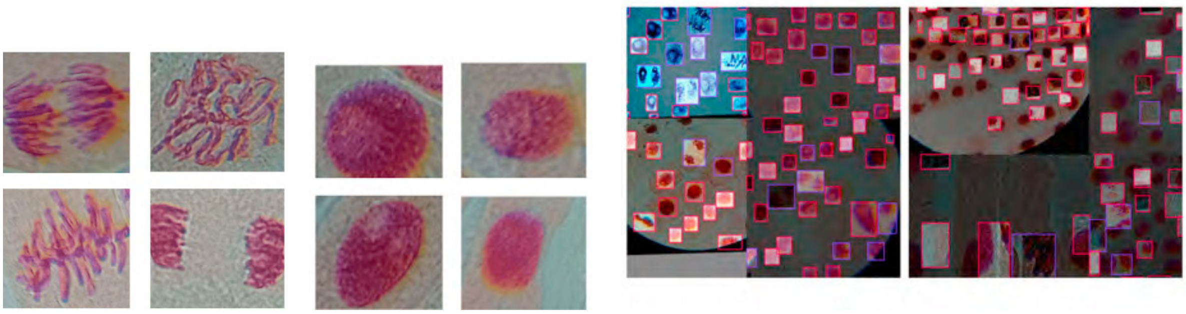


(a) Before preprocessing (b) After preprocessing (c) Sample image annotation

Fig. 2: Data Preprocessing and Annotation

2.3. Data Annotation

The entire image annotation process was conducted using Roboflow. Each image was meticulously annotated, using *d* to indicate dividing cells and *nd* for non-dividing cells as shown in Figures 3a and 3b. The primary focus was on ensuring precision in the delineation of bounding boxes, carefully aligning them to solely encapsulate the nucleus while deliberately excluding any background or cell membrane. This process was aimed to maintain detail and precision in the annotated images. Figure 2c shows an image undergoing processing and annotation.



(a) Dividing Cells (b) Non-Dividing Cells (c) Images after augmentation

Fig. 3: Categorization of data into 2 classes and sample image augmentations

2.4. Dataset Construction and Augmentations

The dataset was divided into three parts: 70% for training, 20% for validation, and 10% for testing. Image augmentations, as shown in Figure 3c, and transformations outlined in Table 1, were applied to the training set, resulting in an increased number of images. The final dataset contained 1548 images, with 1356 images for the training set, 129 images for the validation set, and 63 images for the test set.

2.5. Model Training, Validation, and Testing

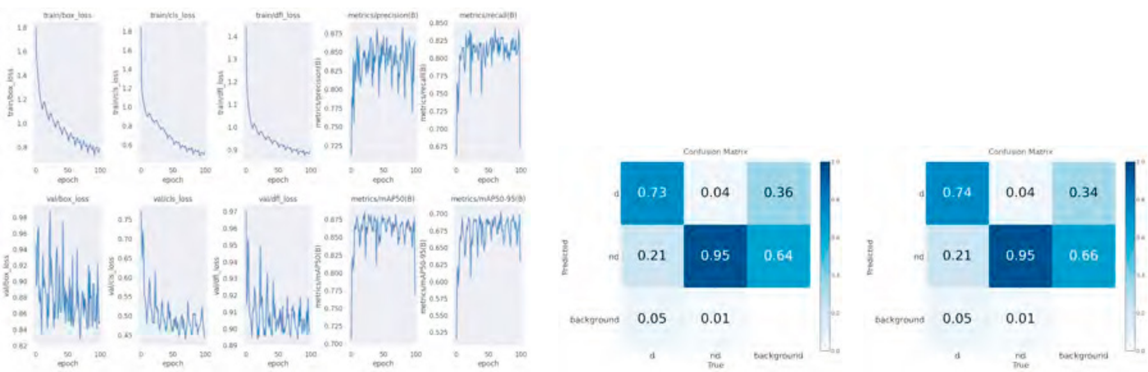
The You Only Look Once (YOLO), represents a series of models renowned for their ability to predict all objects within an image in a single forward pass. These models are categorized as one-stage object detection models, pro-

Table 1: Transformations and Ranges/Descriptions

Transformation	Range/Description
Flip	Horizontal, Vertical
Crop	0% Minimum Zoom, 20% Maximum Zoom
Shear	±15° Horizontal, ±15° Vertical
Saturation	Between −25% and +25%
Brightness	Between −25% and +25%
Noise	Up to 5% of pixels
Mosaic	Applied
Bounding Box Flip	Horizontal, Vertical
Bounding Box Brightness	Between −25% and +25%
Bounding Box Exposure	Between −25% and +25%

cessing entire images through a convolutional neural network (CNN) in a singular forward pass. The YOLOv8 model provides support for custom dataset training, allowing users to tailor the model's learning to specific datasets. The dataset, housed within Roboflow, was formatted in YOLOv8 before integration into a Jupyter notebook for the training process. The training parameters included 100 epochs, a batch size of 16, the utilization of stochastic gradient descent as the optimizer, and a fixed learning rate of 0.01. The strategy employed involved a segmented approach, training the model in 10 iterations of 10 epochs each.

3. Results and Discussion



(a) Training and Validation Loss Graphs (b) Training Confusion Matrix (c) Validation Confusion Matrix

Fig. 4: Performance Metrics Graphs

The results in Figure 4a were obtained after training the model for 100 epochs. The performance metrics for the model resulted in a mean average precision (mAP) score of 88.3%, indicating a high level of accuracy in object detection. Precision stands at 85.9% which reflects the model's capability to make accurate positive predictions among its detections. Furthermore, the recall metric is at 83.0%, showing the model's effectiveness in capturing a substantial proportion of the relevant instances within the dataset.

In the training set, the box loss starts at 1.7998 in epoch 1 and gradually decreases to 0.79343 by epoch 100. The lowest loss is observed at epoch 92. On the other hand, the validation set exhibits a different trend, with the loss starting at 0.94739 in epoch 1 and slightly decreasing to 0.8497 by epoch 100. The highest validation loss is encountered at epoch 23, while the lowest is at epoch 70.

The divergence in trends between the training and validation losses indicates differences in how the model performs on data it has been trained on versus data it has not seen before. As the model continues to train, the box loss on the training set consistently decreases, implying enhanced accuracy in localizing objects. However, the validation loss shows fluctuations, which could imply several things, including overfitting or changes in the complexity of the validation data over epochs.

The classification loss in the training set initiates at 1.8378 during epoch 1 and steadily decreases to 0.52839 by epoch 100. The lowest classification loss is recorded at epoch 92. The loss for the validation set commences at 0.77124 in epoch 1 and shows a slight reduction to 0.44361 by epoch 100. The highest validation loss is encountered in the first epoch, while the lowest occurs at epoch 70.

Within the training set, the distribution focal (DF) loss begins at 1.4371 in epoch 1 and gradually diminishes to 0.89476 by epoch 100. The lowest DF loss is noted at epoch 92. Shifting focus to the validation set, the DF loss initiates at 0.97012 during epoch 1 and experiences a marginal decrease to 0.89801 by epoch 100. The lowest DF loss is identified at epoch 81.

The *metrics/precision(B)* values showcase the evolution of precision throughout the training of the YOLOv8 model. Starting at 0.71157 in the first epoch, the precision steadily increases, reaching 0.86418 by epoch 100. The highest precision of 0.88234 is attained at epoch 81, underscoring a significant improvement in the model's ability to make accurate positive predictions.

The *metrics/recall(B)* values present a snapshot of recall performance across epochs during the training of the YOLOv8 model. Commencing at 0.66182 in the initial epoch, the recall metric sees a modest increase to 0.67451 by epoch 100. The highest recall score of 0.8431 is achieved at epoch 61, indicating improvement in the model's ability to capture relevant instances of the target class.

The *metrics/mAP50(B)* values offer insights into the mean average precision at an IoU (Intersection over Union) threshold of 0.5 throughout the training of a YOLOv8 model. The journey begins at an mAP50 of 0.70589 in the first epoch, experiencing an upward trajectory to 0.76985 by epoch 100. The peak mAP50 value of 0.88665 is achieved at epoch 22, reflecting a substantial improvement in the model's overall precision and recall at the specified IoU threshold.

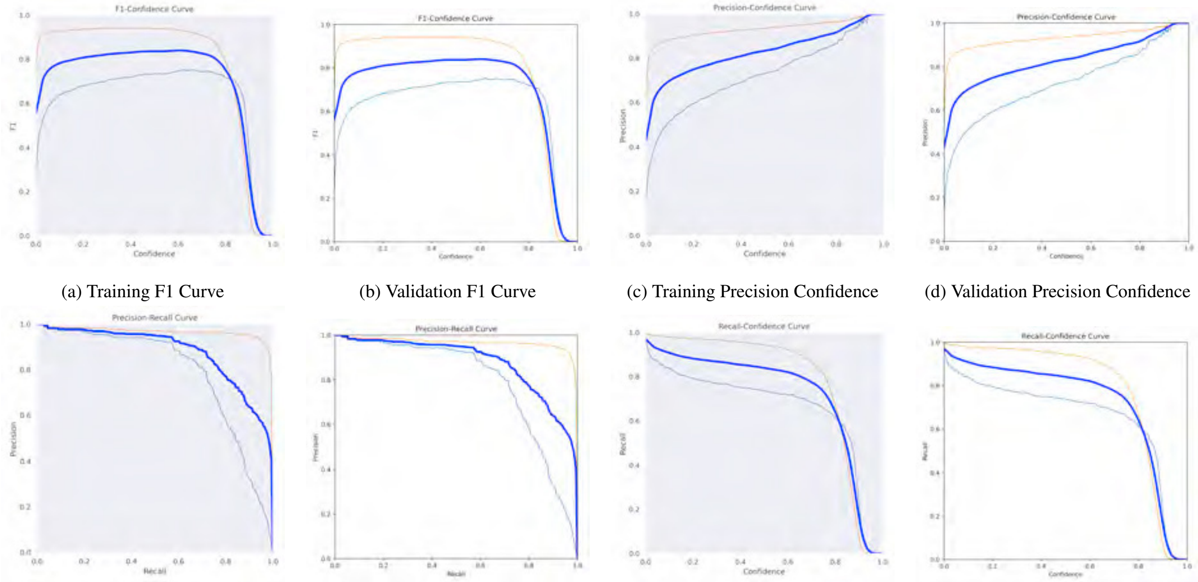
Simultaneously, the *metrics/mAP50-95(B)* values provide a broader perspective, considering the mean average precision across a range of IoU thresholds from 0.5 to 0.95. Initiated at 0.51457 in the first epoch, the mAP50-95 increases to 0.62905 by epoch 100. The highest mAP50-95 value of 0.70351 is reached at epoch 70, signifying an improvement in the model's ability to detect and localize objects across varying IoU thresholds.

The detailed examination of loss per epoch in both training and validation sets reveals distinctive patterns. The gradual decrease in loss values indicates the model's learning capability over epochs, albeit the occasional fluctuation, reflecting periods of enhanced learning and fine-tuning. These patterns suggest that the model's performance improvements are iterative and non-consistently incremental, potentially indicating areas for further optimization. Moreover, the model's precision, recall, and mean average precision (mAP) scores portray its proficiency in making accurate positive predictions among detections and capturing relevant instances within the dataset.

Analyzing the model's classification performance may be acquired by examining the confusion matrices of the training and validation sets observed in Figures 4b and 4c. In both sets, the model exhibits impressive true positive rates for the dividing (*d*) and non-dividing (*nd*) classes, scoring 73% and 95% in the training set and 74% and 95% in the validation set, respectively. Nevertheless, misclassification poses difficulties: 4% of true *nd* cases are incorrectly predicted as *d* in both the training and validation sets, whereas true *d* instances are misclassified as *nd* at a rate of 21% in both sets.

The confusion matrices of both training and validation sets offer insights into the model's classification abilities. While achieving commendable true positive rates for both classes (*d* and *nd*), the misclassification rates suggest areas where the model struggles, particularly in distinguishing between certain classes. These findings emphasize the need for further refinement to enhance the model's discriminative capabilities.

According to the graph of the training set shown in Figure 5a, the *d* class steadily rises to a plateau near 0.8, holds steady at this point, then declines quickly as confidence level approaches 1.0. Similar trends are shown in the *nd* class, which has a higher starting F1 score, a plateau above *d*, and a decline at higher confidence levels. At a confidence level of 0.605, the collective class performance peaks at an F1 value of 0.84. As observed in the graph of the validation set shown in Figure 5b, *all classes* F1 score reaches its maximum 0.84 at a confidence level of around 0.603. The *nd* class continuously maintains a higher F1 score than the *d* class across all confidence levels, highlighting its superior performance in the mean of precision and recall, which is also the case for the training set. The curves for the *d* and *nd* classes show similar patterns.



(a) Training F1 Curve (b) Validation F1 Curve (c) Training Precision Confidence (d) Validation Precision Confidence

(e) Training Precision Recall (f) Validation Precision Recall (g) Training Recall Confidence (h) Validation Recall Confidence

Fig. 5: Performance Metrics Graphs

As observed in Figures 5c and 5d, the training set, as confidence level rises, the *nd* class exhibits increasing precision, surpassing the *d* class consistently. The *all classes* curve achieves precision of 1.00 at approximately 0.956 confidence. In the validation set, the *nd* class shows a rapid initial rise in precision, leveling off with increasing confidence. The *d* class precision increases more gradually than *nd*, reaching values lower than the *nd* class. The *all classes* precision increases as confidence level increases, then later achieving precision (1.00) at around 0.954 confidence.

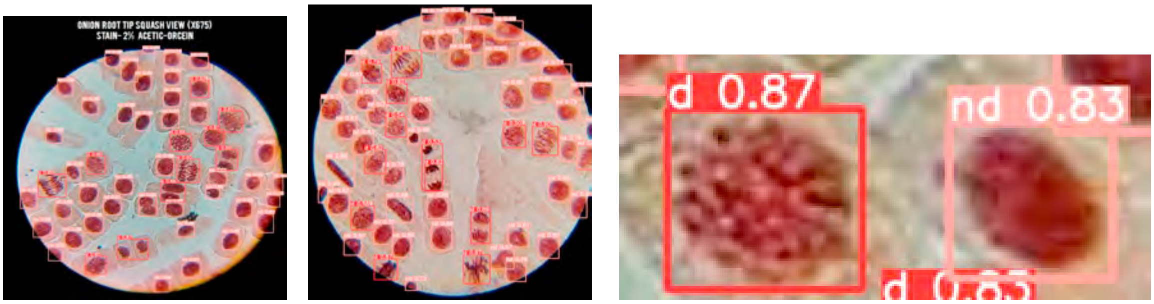
For both training and validation set, the graphs show that the maximum level of combined precision holds true for all higher confidence levels. The graphs suggest that as the confidence level is raised, the model's overall precision improves dramatically, indicating that the more confident the model is in its predictions, the more precise it is.

For both the training and validation sets, the graphs in Figures 5e and 5f show consistent patterns. The *d* class begins with high precision but shows a more rapid decline as recall increases, while the *nd* class maintains high precision until the recall gets closer to 1.0 and rapidly decreases. An overall picture is given by the *all classes* curve, which starts below 1 and gets smaller as recall increases. In addition, the mean average precision (mAP) score for the *nd* class at a confidence level of 0.5 is 0.963 for the training set and 0.964 for the validation set, indicating a little variation in performance between the two sets. The remaining mean average precision (mAP) values for *d* and *all classes* are 0.792 and 0.878 which is the same for both training and validation. The *nd* class also shows the greatest mAP in both sets, followed by *all classes* and *d*, respectively.

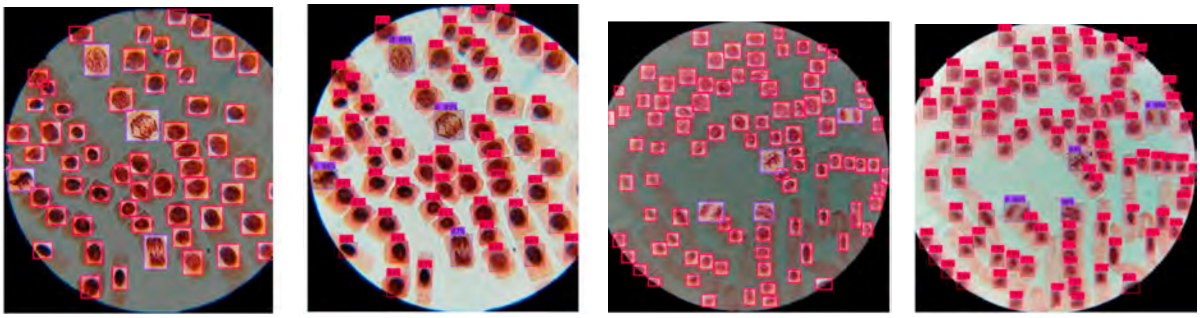
As shown for both the training set and validation set in Figures 5g and 5h, different patterns for the *nd* and *d* classes can be observed. A slower drop in recall as confidence levels rise is indicated by the *nd* class's smoother slope. On the other hand, the *d* class exhibits a quicker fall, indicating a more significant recall decrease with increasing confidence levels. For both training set and validation set's recall-confidence curve for *all classes*, recall is highest with low confidence levels and gradually decreases with increasing confidence levels, even at the lowest confidence level of 0.000, where it is still 0.97.

For both training and validation sets, the overall pattern of declining recall with increasing confidence levels is consistent; however, individual recall values can vary, indicating possible variations in model performance.

The model's efficacy is stronger in analyzing static images depicted in Figures 6a and 6b than in real-time camera detections, encountering challenges with varying environmental conditions like inconsistent lighting during live



(a) Sample Test Set A (b) Sample Test Set B (c) Dividing and non-dividing cells in *prophase* and *interphase* stages.



(d) Sample Actual A (e) Sample Predicted A (f) Sample Actual B (g) Sample Predicted B

Fig. 6: Classifications and Predictions

detections. Sensitivity to data variability is evident, impacting the model's ability to detect instances within images, especially when instances are numerous, leading to difficulties in recognition.

In practical scenarios, the model exhibits commendable performance in detecting and categorizing cells, particularly noticeable when the number of cells in the input image is low. However, as the number of cells within the input image increases, the model's accuracy becomes unstable, especially when it comes to detecting cells. For instance, in Figure 6e, the model successfully classified all detected cells, distinguishing between 4 dividing cells and the remaining non-dividing cells. However, it missed detecting a single cell, resulting in the identification of only 54 non-dividing cells instead of the actual count of 55. This pattern persists in Figure 6g where the model failed to detect an additional 2 cells.

In Figure 6c, the model demonstrates proficiency in distinguishing between the *prophase* and *interphase* stages as dividing and non-dividing cells.

An effective performance in distinguishing between dividing and non-dividing cells is observed within a reasonable range of cell instances. However, the quality of input images plays a crucial role in determining the performance of the model. Higher-resolution images with minimized noise contribute to enhancing the accuracy and precision of cell detection. By providing clearer visual information, these enhanced images enable more accurate identification and classification of cells, ultimately improving the overall performance of the model.

4. Conclusion

In this study, a model based on YOLOv8 was used to detect and classify onion cells in various mitotic stages. The cells were categorized as either dividing or non-dividing. A dataset of 1356 high-quality images of squashed and stained onion root tip cells was used to train the model. The training spanned 500 epochs, halting early due to decreased precision observed between epochs 101 to 500. The model achieved a mAP of 88.3%, a precision of 85.9%, and a recall of 83.0%. However, it faced difficulties in accurately identifying every instance within images containing a high cell count. Furthermore, occasional misclassifications occurred, particularly in specific cell stages. It also demonstrated proficiency in distinguishing between the *prophase* and *interphase* stages as dividing and non-dividing cells, and displayed improved accuracy when presented with high-quality images. Its effectiveness is more pronounced in analyzing static images compared to real-time camera detections, where environmental variations like inconsistent lighting pose challenges.

Experiment results show that the proposed approach of quantifying dividing and non-dividing onion cells can be an alternative method to expedite the calculation of the mitotic index in laboratory experiments that employ the Allium cepa mitotic test in the search for novel anticancer drugs.

Further investigation may involve refining the model, improving and extending the dataset, conducting extensive validation, and exploring potential user interface development.

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